Example 2: Centroid Linkage with Euclidean Distance

Question. For the one-dimensional data set **{7, 10, 20, 28, 35}**, perform agglomerative clustering with centroid linkage and plot the dendrogram to visualize it using Euclidean distance.

EUCLIDEAN DISTANCE

$d(x, y) = \sqrt{\Sigma}\ (x_i - y_i)^2$. In one dimension this is simply **$d(x, y) = |x - y|$**.

CENTROID LINKAGE

The distance between two clusters is the distance between their **centroids (means)**. After each merge, the new cluster is represented by the mean of all its points.

Procedure: start with each value as its own cluster, merge the two closest clusters, compute the new centroid and update distances between centroids, and repeat until one cluster remains.

Solution

STEP 0 · DISTANCE MATRIX

	7	10	20	28	35
7	0				
10	3	0			
20	13	10	0		
28	21	18	8	0	
35	28	25	15	7	0

```
d(7,10)    √(7 − 10)² = 3
d(7,20)    √(7 − 20)² = 13
d(7,28)    √(7 − 28)² = 21
d(7,35)    √(7 − 35)² = 28
d(10,20)   √(10 − 20)² = 10
d(10,28)   √(10 − 28)² = 18
d(10,35)   √(10 − 35)² = 25
d(20,28)   √(20 − 28)² = 8
d(20,35)   √(20 − 35)² = 15
d(28,35)   √(28 − 35)² = 7
```

STEP 1

	7	10	20	28	35
7	0				
10	**3**	0			
20	13	10	0		
28	21	18	8	0	
35	28	25	15	7	0

Smallest entry = **3** between 7 and 10. Merge them into **{7,10}** at height 3.

Update distances (centroid linkage = distance between means):

```
d({7,10}, 20)   centroid({7,10}) = 8.50, centroid(20) = 20 → |8.50 − 20| = 11.50
d({7,10}, 28)   centroid({7,10}) = 8.50, centroid(28) = 28 → |8.50 − 28| = 19.50
d({7,10}, 35)   centroid({7,10}) = 8.50, centroid(35) = 35 → |8.50 − 35| = 26.50
```

STEP 2

	{7,10}	20	28	35
{7,10}	0			
20	11.50	0		
28	19.50	8	0	
35	26.50	15	**7**	0

Smallest entry = **7** between 28 and 35. Merge them into **{28,35}** at height 7.

Update distances (centroid linkage = distance between means):

```
d({28,35}, {7,10})  centroid({28,35}) = 31.50, centroid({7,10}) = 8.50 → |31.50 − 8.50| = 23
d({28,35}, 20)      centroid({28,35}) = 31.50, centroid(20) = 20 → |31.50 − 20| = 11.50
```

STEP 3

	{7,10}	20	{28,35}
{7,10}	0		
20	11.50	0	
{28,35}	23	**11.50**	0

Tie: {7,10}–20 and 20–{28,35} are both 11.5, because 20 lies exactly halfway between the centroids 8.5 and 31.5. Either merge is valid; this solution follows the given dendrogram (SciPy), which joins 20 with {28,35}.

Smallest entry = **11.50** between 20 and {28,35}. Merge them into **{20,28,35}** at height 11.50.

Update distances (centroid linkage = distance between means):

`d({20,28,35}, {7,10})` `centroid({20,28,35}) = 27.67, centroid({7,10}) = 8.50 → |27.67 − 8.50| = 19.17`

STEP 4

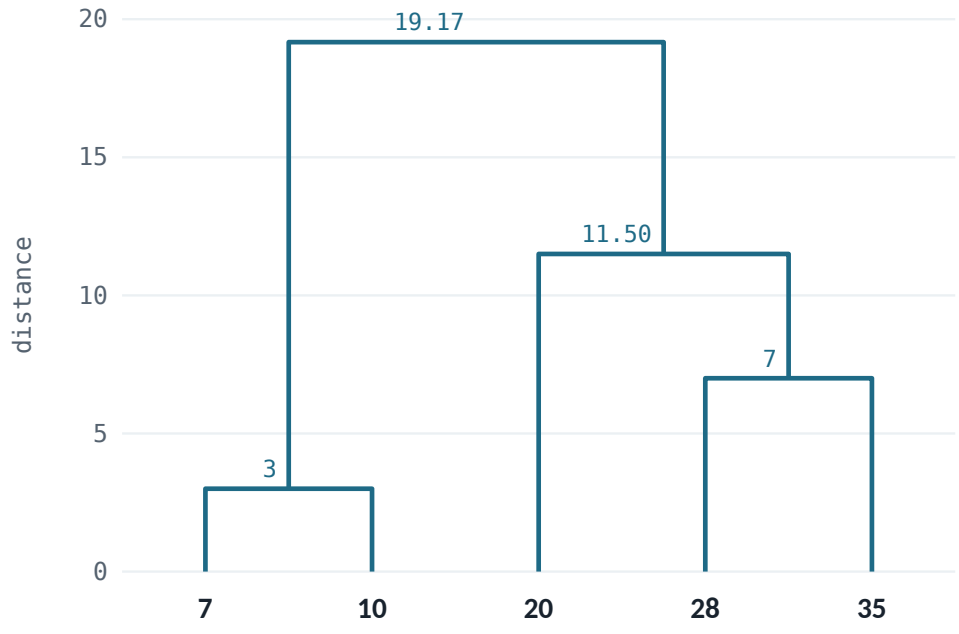
	{7,10}	{20,28,35}
{7,10}	0	
{20,28,35}	**19.17**	0

Smallest entry = **19.17** between {7,10} and {20,28,35}. Merge them into **{7,10,20,28,35}** at height 19.17.

Only one cluster remains. Clustering is complete.

Summary of merges

MERGE	CLUSTERS JOINED	HEIGHT
1	7 + 10	3
2	28 + 35	7
3	20 + {28,35}	11.50
4	{7,10} + {20,28,35}	19.17



Dendrogram · centroid linkage, Euclidean distance

Centroids and interpretation

STEP	MERGED	NEW CENTROID	HEIGHT
1	7 + 10	(7+10)/2 = 8.5	3
2	28 + 35	(28+35)/2 = 31.5	7
3	20 + {28,35}	(20+28+35)/3 = 27.67	11.5
4	{7,10} + {20,28,35}	(7+10+20+28+35)/5 = 20	19.17

- A cut between heights 11.5 and 19.17 gives **2 clusters**: {7, 10} and {20, 28, 35}.
- Compared with single linkage (Example 1), the tree shape is the same but 20 joins at 11.5 instead of 8, and the root is at 19.17 instead of 10. Centroid linkage measures to the centre of {28, 35} (31.5), not to its nearest point (28).
- In one dimension, City block and Euclidean distance are identical, so the differences come only from the linkage rule.
- With the other tie-break ({7,10} + 20 at 11.5, centroid 12.33), the root is again at $|31.5 - 12.33| = 19.17$, but the tree shape differs. Always state which tie-break you used.